Ketamine

(Ketalar)

I. Classification:

- A non-barbiturate dissociative anesthetic drug

II. Actions:

- Produces dream like state – dissociative anesthesia
- Has analgesic (pain control) as well as sedative effects
- Works without causing loss of airway control
- Has limited respiratory or cardiovascular effects, and is less likely to cause hypotension or apnea than benzodiazepines or etomidate
- Ketamine is known for having a wide margin of safety, typically provides good pain relief or sedation, and is generally well tolerated

III. Indications:

- Sedation for RSI (credentialed paramedics only)
- Pain control
- Excited Delirium
- Sedation Intubated patients
- Severe Respiratory Distress from Asthma/COPD
- Adult Cardiac Arrest possibly caused by Asthma Exacerbation
- Refractory Seizures

IV. Contraindications:

- Allergy to drug

V. Precautions:

- Must give SLOWLY (over 60 seconds). Rapid administration can cause respiratory depression or apnea (Use ketamine in 100 mL IVPB wide open)
- Must monitor very closely after administration – place cardiac monitor and respiratory monitor on the patient as soon after administration as possible
- Safety for use in pregnancy is not well established.

VI. Drug interactions:

- Multiple including significant interactions below
  - Use with other central nervous system depressants can cause increased sedation.
  - SSRIs- may increase toxic effect of SSRIs

VII. Adult dosing:

- For **Adjunct Pain Control**
  - 0.2 mg/kg IV/IO/IM over 60 seconds, may repeat x 1 if needed
- For **Severe Pain, Severe Burns**
  - 1-2 mg/kg IV/IO over 60 seconds or 2-4 mg/kg IM, may repeat x1 if needed
- For **Post Intubation Sedation**
  - 1-2 mg/kg IV/IO over 60 seconds every 10-20 minutes
  - 0.1-0.5 mg/kg bolus followed by 0.2-2.5 mg/kg/hr infusion
- For **Severe Respiratory Distress caused by Asthma/COPD**
  - 0.2 mg/kg IV over 60 seconds, may repeat x 1 after 10 minutes if needed
- For **Cardiac Arrest possibly caused by Asthma Exacerbation**
  - 2 mg/kg IV/IO

- For **Extreme Agitation**
  - 4 mg/kg IM (max 2 ml each IM injection Deltoid and 5 mL Thigh)

VIII. Pediatric dosing:

- For **Adjunct Pain Control** (age 8 years old and greater)
  - 0.2 mg/kg IV over 60 seconds, may repeat x 1 if needed
- For **Severe Pain, Severe Burns**
  - 1 mg/kg IV/IO/ over 60 seconds or 2 mg/kg IM, may repeat x1 if needed
- For **Post Intubation Sedation** (age 1 year or greater)
  - 2 mg/kg IV/IO over 60 seconds every 10-20 minutes
- For **Severe Respiratory Distress caused by Asthma/COPD** (age 8 years and greater)
  - 0.2 mg/kg IV over 60 seconds, may repeat x 1 after 10 minutes if needed
- For **Severe Respiratory Distress** caused by Croup or Bronchiolitis
  - Consult Medical Director for 0.2 mg/kg IV/IO
- For **Cardiac Arrest possibly caused by Asthma Exacerbation**
  - 2 mg/kg IV/IO
- For **Extreme Agitation**
  - 4 mg/kg IM (max 1 mL each IM injection Deltoid and 2 mL Thigh)

IX. Maximum doses

- General pain doses: 25mg
- Burn pain or SEVERE trauma pain doses: 500mg IM or 250mg IV
- Sedation doses: 500mg IM or 250mg IV

X. Onset:

- 30 seconds IV
- 3-4 minutes IM

XI. Duration:

- 5-10 minutes IV
- 10-25 minutes IM
- Sedation is followed by a period of gradual awakening which may last up to an hour or more, depending on the patient. This tends to be longer if given IM.

XII. Possible Adverse Effects:

- Rapid horizontal/Vertical involuntary eye movements (Nystagmus) is expected and can be used as indication of drug effect
- Laryngospasm – typically transient, overcome with BVM if necessary
- Increased salivation / bronchial secretions
- Bradycardia/tachycardia
- Hypertension, usually transient
- As with any sedative, there is a risk of aspiration, though this should be theoretically less with Ketamine since laryngeal reflex remains intact
- Respiratory depression or even apnea may occur, particularly with rapid administration of the drug, so monitor end tidal and pulse ox
- Emergence phenomenon – As patient wakes up from dissociative state, they may become agitated or frightened. May have associated delirium. As drug is metabolized, this should resolve, but severe cases may require a benzodiazepine be given. (Consult medical direction first)

Notes:

- *** Giving a dose too fast may cause respiratory depression*
- *** Do not mix benzodiazepines in same syringe as ketamine*
- *** Treat emergence reaction with benzodiazepine, call medical direction first*
- *** Ketamine may be particularly good for RSI in patients with bronchoconstriction secondary to asthma due to bronchodilator effects of Ketamine*